



The instructor approval is done now make a plan and check wjth basic and simple language yet concisely

Here's your plan, broken into phases. Follow the order — don't skip ahead.

Phase 1: Data Setup (Day 1)

1. Load WildGuardMix (allenai/wildguardmix) — train split has ~87k rows.
2. Create the 4-class label from two columns:
 - `adversarial = False + prompt_harm_label = unharmed` → **Benign-Direct**
 - `adversarial = True + unharmed` → **Benign-Adversarial**
 - `adversarial = False + harmful` → **Harmful-Direct**
 - `adversarial = True + harmful` → **Harmful-Jailbreak**
3. Drop rows where `prompt_harm_label` is null. Keep only the prompt text + your new label.
4. Save as CSV. Done — no more data work needed later.

Phase 2: EDA (Day 1–2)

- Count rows per class (expect imbalance — note it).
- Bar chart of class distribution.
- Prompt length histogram.
- Word cloud per class.
- Check duplicates and empty prompts, remove them.
- Write down 3–5 observations — you'll quote these in the report later.

Phase 3: Preprocessing + Split (Day 2)

- Lowercase, remove URLs, fix whitespace.
- **Don't over-clean** — jailbreak prompts use weird formatting, and that's useful signal. Note this decision in writing.
- Split: stratified 70% train / 15% val / 15% test, `random_state=42`.

Phase 4: Representations (Day 3)

- TF-IDF → for the 3 ML models.
- Word2Vec or GloVe → for the 6 RNN models.
- BERT uses its own tokenizer — no separate representation needed.

Phase 5: Train Models (Day 3–7, the bulk of the work)

Order matters — easiest first:

#	Model	Input	Rough effort
1	Naive Bayes	TF-IDF	30 min
2	Logistic Regression	TF-IDF	30 min
3	Random Forest	TF-IDF	1 hr
4–6	SimpleRNN, GRU, LSTM	Word2Vec/GloVe	1 day (share 90% of code)
7–9	Bi-SimpleRNN, Bi-GRU, Bi-LSTM	Word2Vec/GloVe	half day (one-line changes)
10	BERT Base	BERT tokenizer	1 day (needs GPU — use Colab/Kaggle)

For every model: train 3 different hyperparameter configs, pick the best by validation F1. Log every run in one table (columns: model, config, val F1). Start this table on Day 3, never at the end.

Phase 6: Evaluation (Day 8)

For each model's best config, on the test set:

- Accuracy, F1, confusion matrix heatmap, full classification report.
- One final comparison table of all 10 models.
- Write 2–3 sentences each on best and worst model and why.

Phase 7: Deliverables (Day 9–11)

- **Notebook:** clean structure, markdown headings, all outputs visible, embedded tables/plots.
- **Report** (7–8 pages, ACL template): fill sections in this order — Methodology, Results, EDA, then Intro, Abstract, Related Work last.
- **Presentation:** 8–12 min screen-share recording walking through the code.

Bonus (only if everything above is done)

- GitHub repo, or deploy best model on HuggingFace Spaces / Vercel. Skip the ensemble unless time remains.

Daily Rules

1. Code first, read second. Minimum 2 hours coding before opening any paper.
2. One bug/question → one search → back to code. No "quick background reading."
3. Every experiment gets logged the moment it finishes.

Today's only job: Phase 1 + a TF-IDF Logistic Regression baseline. If you have one number on screen by tonight, the project is real and the rest is iteration.